

Jeffrey Ding

Data Scientist 647-928-5709 | *Email* | *LinkedIn* | *GitHub*

TECHNICAL SKILLS

Machine Learning & AI: Machine Learning, Deep Learning, Predictive Modeling, CNN, SVC, RAG, Agentic Workflows, Prompt Engineering

Programming & Databases: Python, SQL, PostgreSQL, IBM Db2, MongoDB, JavaScript, R

Data Analysis & Modeling: EDA, Data Mining, Data Visualization, Data Warehousing, Data Modelling

Tools & Frameworks: AWS, Git, CI/CD, TensorFlow, Keras, Streamlit, Jupyter, Folium, SPSS, Excel

PROJECTS

Maternal Health Risk Classifier

- Engineered a Support Vector Classifier (SVC) to predict risk tiers using processed IoT data, achieving 77% overall accuracy and 91% recall (0.943 AUC) for critical identification of at-risk subjects.

Detecting Diabetic Retinopathy with CNNs

- Preprocessed and analyzed clinical data utilizing a Convolutional Neural Network (CNN) and Residual blocks using Keras and TensorFlow 2.0 to improve early detection.

Filesystem Organizing API

- Co-authored a Python API utilizing Git, CI/CD, and test-driven development to visualize data and identify bloat, guiding user management.

Interactive Dashboards & Web Apps

- NYC Motor Vehicle Collision Heatmap:** Designed an interactive web app with Python, Folium, and Streamlit to visualize traffic accident data.
- Stock Exchange Revenue Dashboard:** Published a Jupyter notebook that visualizes time-series financial data using dynamic figures.

EXPERIENCE

University of Toronto

Machine Learning for Life Science Mentorship | Jun 2024 - Jul 2025

- Applied learned machine learning and deep learning concepts towards developing independent projects.
- Partnered with researchers as a volunteer clinical research coordinator consultant.

Sunnybrook Research Institute

Data Manager | Feb 2022 - Feb 2024

- Increased trial participation by an estimated 40% by aggregating and screening new patient data.
- Abstracted unstructured data from 250+ patients to evaluate wellness and determine treatment efficacy.
- Orchestrated the simultaneous conduct of 15+ clinical trials by coordinating cross-functional teams.
- Accelerated the activation of 10+ new trials through financial and legal submissions.
- Liaised with sponsor clients by facilitating on-site/remote visits and reconciling 200+ queries.

University Health Network (UHN)

Summer Undergraduate Researcher | May 2019 - Aug 2019

- Performed regression on open-source data to correlate comorbidities and clinical trial participation.
- Investigated provincial healthcare data to optimize the transition of survivors to primary care.
- Facilitated 15 research studies by directing patients and maintaining databases with Excel and SPSS.

EDUCATION

University of British Columbia | *Master of Data Science* | 2025 - 2026

University of Western Ontario | *Bachelor of Medical Sciences* | 2017 - 2021

CERTIFICATIONS

- IBM Data Science Professional Specialization:** IBM, Coursera (2024)